



Meta to acquire a VR company

Meta is acquiring the maker of VR workout app 'Supernatural'.
<https://digit.in/nov21-23>



AI powered drivethroughs at McD

McDonald's and IBM could bring AI-powered drive-thrus to more restaurants.
<https://digit.in/nov21-24>

Colour Gamuts

Throwing some COLOUR onto your displays.

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Colour. It is something that we're used to seeing everyday. The whole essence of having colours in our life is to make it more interesting and vibrant. Colours are there so that things don't get bland and boring. But, have you ever wondered about how your monitor can display them? Sure, the human eye can see billions of colours and those billions of colours are present in nature too. But how does a flat screen, which you stare at everyday, produce almost equal or less colour than mother nature gives us? Or how does a high end camera capture so much colour? What are the standards in capturing and displaying colours? Aren't colours a subjective thing? I mean, a dark orange to you could be perceived as red to someone else. So how do we even bring standardisation in this vastly subjective field? That's exactly the topic for this article and we're sure that we would've answered almost all of those questions by the end of this article.

HOW DO HUMANS PERCEIVE COLOUR?

Well, before we understand modern technology and all the cool stuff, we first need to understand how our eyes perceive colour. When it comes to colour, scientifically, there

is no such thing as "Colour" in the world. It's our teeny tiny brain that perceives the world around us that way. That is exactly why some people are colourblind. Their brain just cannot perceive colour as well as normal people do. But how does it even identify which colour is which? To understand that, we first need to see the visible spectrum. Electromagnetic waves come in all sorts of wavelengths and frequencies. But the human eye can "see" or is "sensitive to" a very small band of those wavelengths. That small band is called the "visible spectrum" or the spectrum of electromagnetic waves that our eyes can see.

As you can see in the diagram, the electromagnetic spectrum is vast. It's wavelength starts from 10^{-14} m and goes upto 10^4 m. Now, as it is clearly visible in the image, we can "see" or "perceive" a very small part of that spectrum. I.e. Electromagnetic waves whose wavelengths start from 10^{-7} m and go till somewhere around 10^{-4} m can be seen by the human eye and those electromagnetic waves are what we call visible light.

Now comes the most important part. How do our eyes translate the light coming from the world around us to colours? Light travels into the eye to the retina, located on the back of the eye. The retina is covered with millions of light receptive cells called rods and cones. When these cells detect light, they send signals to the brain. Most people have three kinds of cone cells, and every colour stimulates more than one cone. Their combined response produces a unique signal for each colour, and millions of different colours can be distinguished this way. These cells, working in combination with con-

